

Problem

A ladder network has a voltage gain of

$$\mathbf{H}(\omega) = \frac{10}{(1 + j\omega)(10 + j\omega)}$$

Sketch the Bode plots for the gain.

Step-by-step solution

Step 1 of 9

Consider the transfer function of voltage gain.

$$\begin{aligned}\mathbf{H}(\omega) &= \frac{10}{(1 + j\omega)(10 + j\omega)} \\ &= \frac{10}{10(1 + j\omega)\left(1 + \frac{j\omega}{10}\right)} \\ &= \frac{1}{(1 + j\omega)\left(1 + \frac{j\omega}{10}\right)}\end{aligned}$$

Thus, the two break frequencies are: 1 and 10 rad/s.

The value of gain, K is 1.

The magnitude of voltage gain is,

$$|\mathbf{H}(\omega)| = \frac{1}{\left(\sqrt{1 + \omega^2}\right)\left(\sqrt{1 + \left(\frac{\omega}{10}\right)^2}\right)}$$

Step 2 of 9

Calculate the decibel value of the magnitude.

$$\begin{aligned}20 \log |\mathbf{H}(\omega)| &= 20 \log \left(\left(\sqrt{1 + \omega^2} \right) \left(\sqrt{1 + \left(\frac{\omega}{10} \right)^2} \right) \right)^{-1} \\ \mathbf{H}_{dB} &= -20 \log \left(\sqrt{1 + \omega^2} \right) - 20 \log \left(\sqrt{1 + \left(\frac{\omega}{10} \right)^2} \right)\end{aligned}$$

Calculate the phase angle ϕ .

$$\phi = -\tan^{-1} \left(\frac{\omega}{1} \right) - \tan^{-1} \left(\frac{\omega}{10} \right)$$

Step 3 of 9

Calculate the intercept of bode plot with magnitude axis.

$$\begin{aligned}c &= 20 \log K \\&= 20 \log 1 \\&= 0 \text{ dB}\end{aligned}$$

For gain K , the bode plot will be constant.

Draw the magnitude plot for the term $(j\omega + 1)$ as shown in Figure 1.

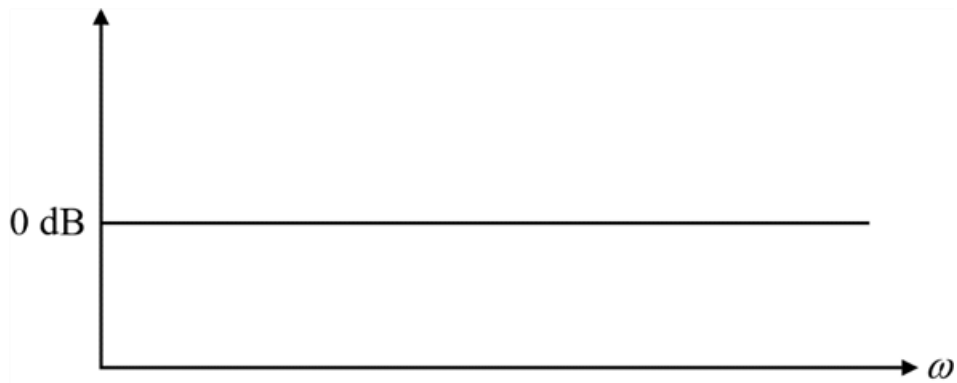


Figure 1

Step 4 of 9

For the term $\frac{1}{(1 + j\omega)}$, the slope will be -20 dB/decade for first order pole starting at

$\omega = 1 \text{ rad/s}$. For $\omega < 1$, the slope for Bode plot of this term will be 0 dB . For $\omega > 1$, the slope for the Bode plot of this term will be -20 dB/decade .

Sketch the Bode plot for $\frac{1}{(1 + j\omega)}$ term.

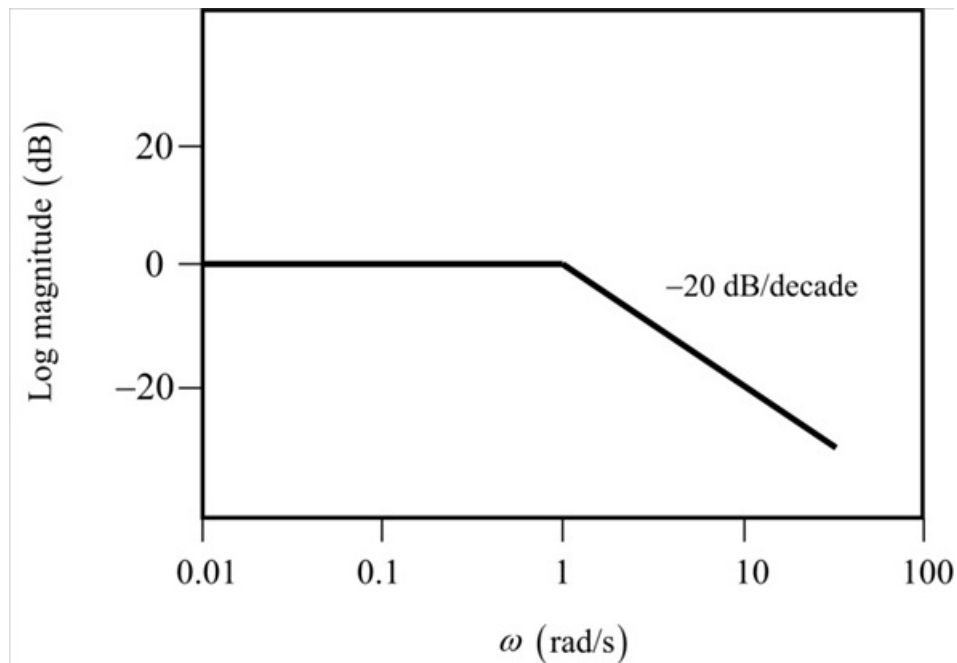


Figure 2

Step 5 of 9

For the term $\frac{1}{(j\omega+10)}$, the slope will be -20 dB/decade for first order pole starting at

$\omega = 10 \text{ rad/s}$. For $\omega < 10$, the slope for Bode plot of this term will be 0 dB . For $\omega > 10$, the slope for the Bode plot of this term will -20 dB/decade .

Sketch the Bode plot for $\frac{1}{(j\omega+10)}$ term.

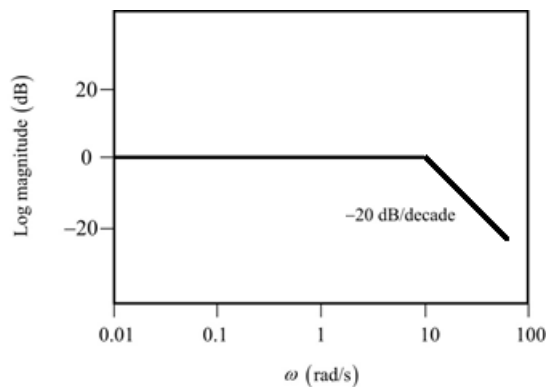


Figure 3

Step 6 of 9

Add the slopes to obtain the Bode plot of the whole transfer function.

Since there are no poles at origin, the starting slope will be 0 dB/decade starting at 0 dB .

For $\omega < 1 \text{ rad/s}$, the slope will remain constant.

At $\omega = 1 \text{ rad/s}$, the slope reduces by -20 dB/decade due to first order pole. Thus, the slope will change from 0 dB to -20 dB/decade .

At $\omega = 10 \text{ rad/s}$, the slope reduces by -20 dB/decade due to quadratic pole. Thus, the slope changes from -20 dB/decade to $(-20 - 20) \text{ dB/decade} = -40 \text{ dB/decade}$.

Step 7 of 9

Calculate the magnitude at $\omega = 10 \text{ rad/s}$.

$$\begin{aligned}
 |H(\omega)| &= \frac{1}{\left(\sqrt{1+\omega^2}\right)\left(\sqrt{1+\left(\frac{\omega}{10}\right)^2}\right)} \\
 &= \frac{1}{\left(\sqrt{1+(10)^2}\right)\left(\sqrt{1+\left(\frac{10}{10}\right)^2}\right)} \\
 &= \frac{1}{\sqrt{101}\sqrt{2}} \\
 &= 0.07036
 \end{aligned}$$

Step 8 of 9

Calculate the magnitude in decibels.

$$\begin{aligned} H_{dB}|_{\omega=10} &= 20 \log |H(j\omega)| \\ &= 20 \log 0.07036 \\ &= -23.05 \text{ dB} \end{aligned}$$

Sketch the magnitude plot is shown in Figure 1.

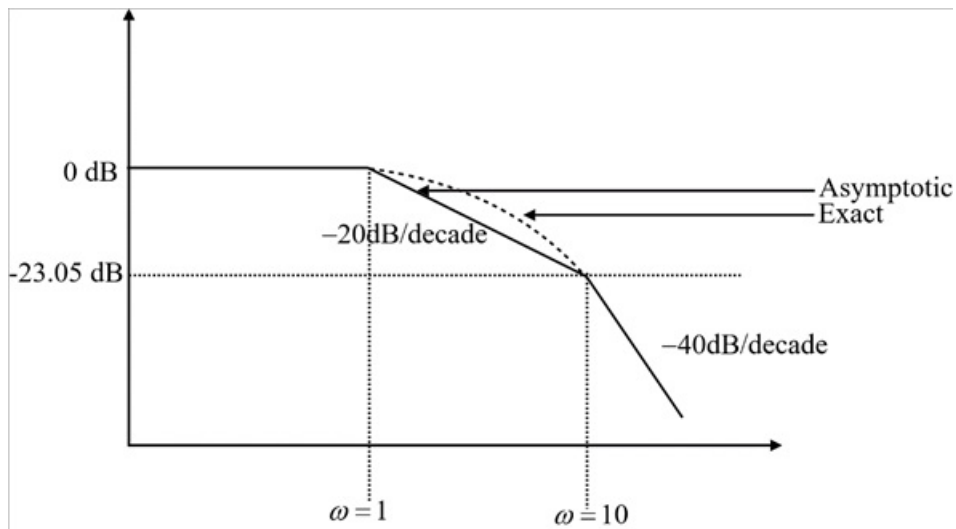


Figure 4

Rewrite the phase equation as:

$$\phi = -\tan^{-1}\left(\frac{\omega}{1}\right) - \tan^{-1}\left(\frac{\omega}{10}\right)$$

Note: Set the radian mode in calculator while calculating the phase and then convert to degrees.

Form a table for value of angle ϕ for different values of ω .

Table1. Phase angle versus ω .

ω (rad/s)	$ G(j\omega) $
0.01	-0.63°
0.1	-6.28°
1	-50.71°
10	-129.289°
100	-173.71°
1000	-179.369°

Step 9 of 9

Sketch the phase plot as shown in Figure 5.

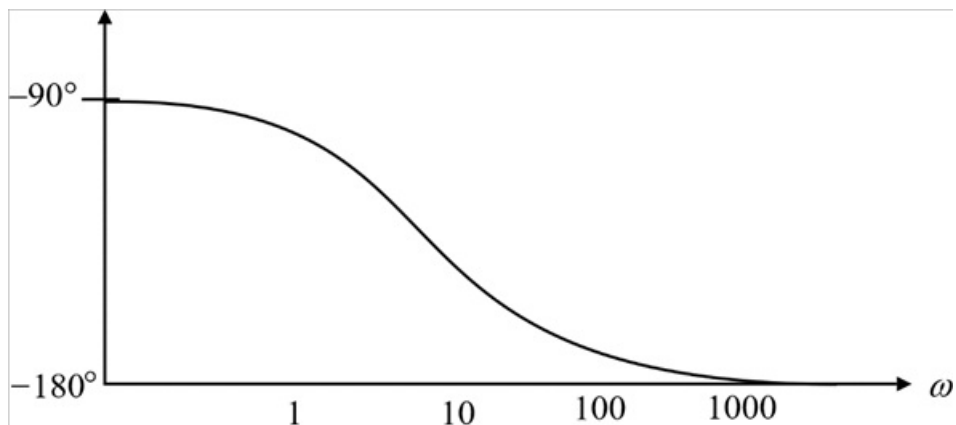


Figure 5

Therefore, the bode plots for the given function are shown in Figure 4 and 5.

